

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Nuclear Physics Lab

L-T-P: 0-0-3

Credits: 2

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

1. To study plateau characteristics of GM detector and determination of recovery time, resolving time and dead time using Oscilloscope .
2. Determination of dead time of GM detector using double source method.
3. Verification of Rutherford scattering formula using Alpha particles on Gold
4. Beta Spectroscopy using magnetic spectrometer using and determination of end-point energy using Fermi-Kurie plot.
5. Alpha Spectroscopy using Si detector .
6. Gamma ray Energy measurement using NaI(Tl) detector (single channel analyser).
7. Determination of electron conversion coefficient using ^{137}Cs source
8. Gamma-Gamma coincidence study using NaI(Tl) detectors
9. Study of Compton-Suppression spectrometer.
10. Quantitative analysis of Natural radioactivity using HPGe detector
11. Measurement of activity of a radioactive source using relative and absolute methods
12. To study the statistical nature of nuclear decay process using GM counter.
13. Determination of linear absorption coefficient of different materials (Z-dependence and Energy-dependence for gamma rays).

Reference Books:

Lab manual